

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA0203

COURSE NAME: C Programming

COURSE OUTCOME COVERED

CO1: Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)

Assessment Tool Weightage: 40%

QUESTIONS:

Total Marks:100

Annexure A

PSEUDO CODE WRITING

Code of Conduct:

I B. Bhanuprakash (192412518) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B. Bhanu prakash

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

1) Pseudocode :-

```
# define DEBUG 1
```

```
IF DEBUG == 1
```

```
    PRINT intermediate_value
```

```
END IF
```

Error :- DEBUG is 0,

2) # define pi 3.14

```
DECLARE radius AS float
```

```
DECLARE area AS float
```

```
INPUT radius
```

```
area = pi * radius * radius
```

```
PRINT area.
```

3) # define RATE 1.5

```
DECLARE x AS int = 10
```

```
DECLARE result AS float
```

```
result = x * Rate
```

```
PRINT result
```

output : 15.0

4) # define Max 1000

```
DECLARE temp AS int
```

```
INPUT temp
```

```
IF temp <= MAX
```

```
    print "valid"
```

```
ELSE
```

```
    print "Invalid"
```

```
END IF
```

5)

1. DEFINE conversion rate

2. INPUT amount

3. calculate converted value

4. PRINT result

6)

define MIN 10

define MAX 50

DECLARE value AS int

INPUT value

IF (value >= MIN AND value <= MAX)

PRINT "within range"

ELSE

PRINT "out of range"

END IF

7) # define DISCOUNT 10

define TAX 5

define ENABLE - DISCOUNT 1

define ENABLE - TAX 1

DECLARE price AS float

DECLARE quantity AS int

DECLARE total AS float

INPUT price

INPUT quantity

total = price * quantity

IF ENABLE - DISCOUNT

total = total - (total * DISCOUNT / 100)

END IF

IF ENABLE - TAX

total = total + (total * TAX / 100)

END IF

PRINT total

8) # define TAX 0.05

DECLARE price AS float

DECLARE total AS float

INPUT price

total = price + (price * TAX)

PRINT total

9) #define MODE 1

DECLARE temp AS float

DECLARE result AS float

INPUT temp

IF MODE == 1

result = (temp * 9/5) + 32

ELSE

result = (temp - 32) * 5/9

END IF

PRINT result

10) #define SIZE 5

DECLARE arr[SIZE]

DECLARE sum AS int = 0

DECLARE average AS float

FOR i = 0 TO SIZE - 1

INPUT arr[i]

sum = sum + arr[i]

END FOR

average = sum / SIZE

PRINT sum

PRINT average.